

Operator-Ball Transfer of Finite-Mesh Schur Certificates

One Denominator for Assembly Error and Contour Interpolation

Prime Dynamics Theory Program

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Abstract

RH-167 reduces continuous contour control to exact sample inverse bounds. Physical computations instead provide a nominal block, an approximate inverse, and an operator uncertainty ball. We combine all three errors in a single Banach-lemma transfer. If $\|T - \hat{T}\| \leq \eta$, every point of a contour cell is within h_k of z_k , and

$$\|(z_k - \hat{T})^{-1}\| \leq m_k, \text{quad} m_k(h_k + \eta) < 1,$$

then

$$\|(z - T)^{-1}\| \leq \frac{m_k}{1 - m_k(h_k + \eta)}.$$

For a fixed packet projection, an operator ball also enlarges each directed coupling by at most its radius. These facts yield a robust finite Schur gate for the unknown exact operator. A dense 256-case audit has no transfer failure. The formulas are rigorous; the audit is not formal interval arithmetic and does not supply physical prime-dynamics balls.

1 Nominal inverse plus operator uncertainty

Let T be the exact block and $\hat{T}$ a nominal approximation with $\|T - \hat{T}\| \leq \eta$. At a mesh node z_k , suppose the nominal resolvent has certified norm at most m_k . For $|z - z_k| \leq h_k$,

$$z - T = (z_k - \hat{T}) \left[I + (z_k - \hat{T})^{-1}((z - z_k)I - (T - \hat{T})) \right],$$

up to the equivalent reversed factorization. The perturbation norm is at most $m_k(h_k + \eta)$.

Theorem 1 (Joint mesh/operator-ball envelope). *If $m_k(h_k + \eta) < 1$ on every contour cell, then $\Gamma \subset \rho(T)$ and*

$$\sup_{z \in \Gamma} \|(z - T)^{-1}\| \leq \max_k \frac{m_k}{1 - m_k(h_k + \eta)}.$$

Proof. The bracketed factor is invertible by a Neumann series, with inverse norm at most $[1 - m_k(h_k + \eta)]^{-1}$. Multiplication by the nominal sample inverse and maximization over cells prove the result [Kato, 1995]. $\square$

The mesh displacement and operator error add before multiplication by the sample condition. Treating them in two successive Neumann transfers is valid but generally loses more margin.

2 Certifying the nominal sample inverse

Let R_k approximate $(z_k - \hat{T})^{-1}$ and suppose

$$\|I - R_k(z_k - \hat{T})\| \leq e_k < 1.$$

Proposition 2 (A-posteriori inverse bound). *The nominal sample is invertible and*

$$\|(z_k - \hat{T})^{-1}\| \leq \frac{\|R_k\|}{1 - e_k}.$$

Proof. Write $R_k(z_k - \hat{T}) = I - E_k$. The right-hand side is invertible by the Neumann series. In finite dimensions a certified right inverse is the inverse, and the norm estimate follows. The same conclusion holds for a two-sided Banach-space approximate inverse with the usual range hypotheses. $\square$

Thus a validated implementation needs only outward bounds for $\|R_k\|$, $\|E_k\|$, the geometric h_k , and the operator radius.

3 Coupling inflation for a fixed packet

Let P be fixed, $Q = I - P$, and assume $\|A - \hat{A}\| \leq \eta$. Then

$$\begin{aligned} \|PAQ\| &\leq \|P\hat{A}Q\| + \eta, \\ \|QAP\| &\leq \|Q\hat{A}P\| + \eta, \end{aligned}$$

while the packet and complement compressions each lie within η of their nominal blocks.

Corollary 3 (Robust finite Schur gate). *Apply Theorem 1 to the two nominal diagonal blocks, obtaining exact contour bounds a and d . If nominal coupling bounds are $\hat{b}, \hat{c}$ and*

$$ad(\hat{b} + \eta)(\hat{c} + \eta) < 1,$$

then the exact operator preserves the enclosed packet rank.

Proof. The compression and coupling inequalities transfer all finite data to the exact blocks. Apply the two-sided Schur theorem [Program, 2026]. $\square$

If the packet projection is itself uncertain, it must first be transported to a fixed realized projection or enclosed with additional centered projector-action bounds. The displayed $+\eta$ rule must not be reused for an untracked moving packet.

4 Structured error budgets

A single full-operator radius is convenient but can be unnecessarily pessimistic. Suppose the validated assembly gives separate radii

$$\eta_P, \quad \eta_Q, \quad \eta_B, \quad \eta_C$$

for the packet block, complement block, return coupling, and outward coupling. Then Theorem 1 uses η_P and η_Q in the two resolvent denominators, while the exact feedback test becomes

$$a(\eta_P)d(\eta_Q)(\widehat{b} + \eta_B)(\widehat{c} + \eta_C) < 1.$$

Proposition 4 (Monotonicity of structured certification). *If every structured radius is at most a common radius η , the structured feedback upper is no larger than the common-radius upper, whenever both denominators close.*

Proof. Each transferred resolvent envelope and each inflated coupling is monotone in its nonnegative radius. Their product is therefore monotone in all four arguments. $\square$

This identifies where validation effort is most valuable: reducing the radius attached to the larger sample resolvent can improve a denominator far more than uniformly adding precision to already benign blocks.

5 Four distinct certificate failures

The robust pipeline has four logically different red outcomes.

1. If a nominal inverse defect is at least one, the selected sample inverse is not certified; better linear algebra or precision may repair it.
2. If $m_k(h_k + \eta_P) \geq 1$ or its complementary analogue, the sample neighborhood is too large for the available conditioning; mesh refinement or a smaller operator ball may repair it.
3. If both contour envelopes close but $adbc \geq 1$, the current Schur feedback corridor is inconclusive; changing only the mesh cannot remove a genuine coupling product.
4. If the rank gate passes but the directional packet-diagonal bound is at least one, spectral multiplicity is certified while the requested graph conditioning remains unproved.

These outcomes should be stored separately. Collapsing them into one Boolean “R failed” would lose the information needed to decide whether to increase precision, change the contour, redesign the packet, or abandon the present graph architecture.

6 Audit and route consequence

The audit samples 256 upper-triangular nominal matrices, adds a perturbation with known numerical operator norm, and builds a 24–64 node circle mesh. It compares the theorem envelope with 2048 dense resolvent evaluations of the perturbed matrix. No accepted case fails. Because ordinary floating norms are used, this is a diagnostic test of the formula rather than an outward-rounded publication certificate.

RH-168 completes the fixed-scale finite certificate architecture in abstract form. What remains is to build the physical ambient packet, generate validated operator and inverse balls, and control the family across scales. No such all-level result, Gate A determinant, self-adjoint operator, zeta identity, or RH conclusion is claimed.

References

Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1995.

Prime Dynamics Theory Program. A two-sided schur certificate for packet-to-riesz lifting, 2026. RH-163 manuscript.